JACON - component power ratings and signal-chain budget

Mini-Circuits parts on hand, with the maximum power each one can take, taken directly from the manufacturer datasheets.

Every number in this document is traceable. Each part has a screenshot cropped from its own datasheet PDF with the page number, plus links to the Mini-Circuits product page and to the PDF saved in datasheets/ next to this file. Nothing here is written from memory: if a value looks wrong, open the linked PDF at the cited page and check it.

Conversion used throughout: P(mW) = 10^(P(dBm)/10). 0 dBm = 1 mW, +17 dBm = 50 mW, +30 dBm = 1 W, -15 dBm = 0.0316 mW.

1. Maximum power each part can take

Two different limits matter. The damage limit is what destroys the part. The P1dB / compression limit is where it stops behaving linearly, which happens long before damage. Both are reproduced as printed on the datasheet, then converted to dBm and mW.

Part	Damage limit, as printed on the datasheet	= dBm	= mW	P1dB / compression	= mW	Page
ZX60-5123LN+	RF Input Power (CW) +15 dBm	+15.00	31.62	+17.0 dBm	50.1 mW	p.3
ZX05-14LH-S+	RF Power 50 mW	+16.99	50	+5.0 dBm	3.2 mW	p.1
ZX60-14012L-S+	Input Power (no Damage) 10 dBm	+10.00	10	+11.0 dBm	12.6 mW	p.1
ZX10-2-1252-S+	Power Input (as a splitter) 1.85 W	+32.67	1850	passive - n/a	-	p.2
ZX60-P162LN+	Input RF Power (No Damage) VD=+4 V: +25 dBm	+25.00	316.2	+19.9 dBm	97.7 mW	p.2
ZX60-P103LN+	Input RF Power (no damage): 21 dBm (50-2000 MHz), 26 dBm (2000-3000 MHz)	+21.00	125.9	+22.4 dBm	173.8 mW	p.3
ZX60-2534MA-S+	Input Power (no damage) -15 dBm	-15.00	0.03162	+19.0 dBm	79.4 mW	p.1
ZFRSC-42-S+	Power Input (as a splitter) 0.75 W	+28.75	750	passive - n/a	-	p.1
ZX05-30W-S+	RF Power 50 mW	+16.99	50	+1.0 dBm	1.3 mW	p.1

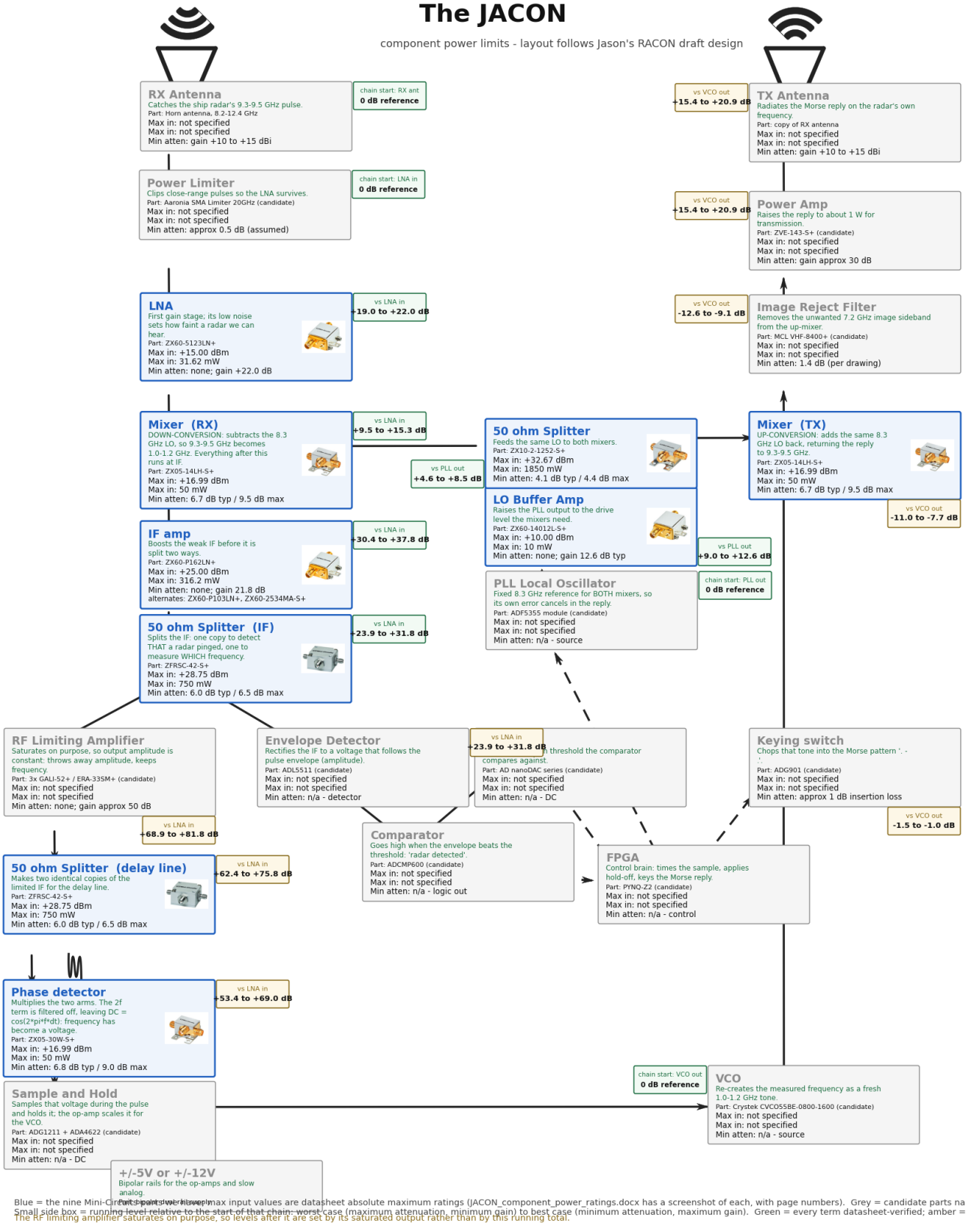
WARNING - the ZX60-2534MA-S+ stands out: its -15 dBm (0.032 mW) damage limit is about 36 dB lower than any other amplifier here. Placed directly after the RX mixer it would be destroyed by anything except a distant echo.

2. Where each part sits, and what it sees

Part	Role	Datasheet band	Frequency it sees in JACON	Gain / loss at that frequency
ZX60-5123LN+	RX LNA	5.5 - 12.5 GHz	9.3 - 9.5 GHz	+22.0 dB typ (19.0 min), 5.5-12.5 GHz
ZX05-14LH-S+	RX down-mixer / TX up-mixer	RF+LO 3.7 - 10 GHz, IF DC - 4 GHz	RF 9.3-9.5 GHz, LO 8.3 GHz, IF 1.0-1.2 GHz	conv. loss 7.72 dB typ at RF 9400 MHz (6.7 dB band typ), LO +10 dBm
ZX60-14012L-S+	LO buffer amplifier	0.3 MHz - 14 GHz	8.3 GHz	12.55 dB at 8.0 GHz, 11.85 dB at 9.0 GHz -> approx 12.3 dB at 8.3 GHz
ZX10-2-1252-S+	LO 2-way splitter (DC pass)	1800 - 12500 MHz	8.3 GHz	-3.0 dB split + 0.7 dB typ insertion loss
ZX60-P162LN+	IF amplifier (preferred for 1.0-1.2 GHz)	0.7 - 1.6 GHz	1.0 - 1.2 GHz	21.81 dB at 1.1 GHz (22.49 at 1.0, 21.12 at 1.2 GHz)
ZX60-P103LN+	IF amplifier (wideband alternate)	50 - 3000 MHz	1.0 - 1.2 GHz	+15.6 dB typ at 1000 MHz (13.5 min)
ZX60-2534MA-S+	IF amplifier (high gain alternate)	0.5 - 2.5 GHz	1.0 - 1.2 GHz	+43.5 dB typ at 1.0 GHz (at VD = 5 V)
ZFRSC-42-S+	Resistive 2-way splitter (IF split and delay-line split)	DC - 4200 MHz	1.0 - 1.2 GHz	-6.0 dB total loss (5.99 / 6.02 dB at 1000 MHz)
ZX05-30W-S+	Phase detector (delay-line	RF+LO 300 - 4000 MHz, IF	RF+LO 1.0-1.2 GHz, IF = DC	conv. loss 6.22 dB typ at RF 944

	mixer)	DC - 950 MHz		MHz (6.8 dB band typ), LO +7 dBm

3. Block diagram with power limits

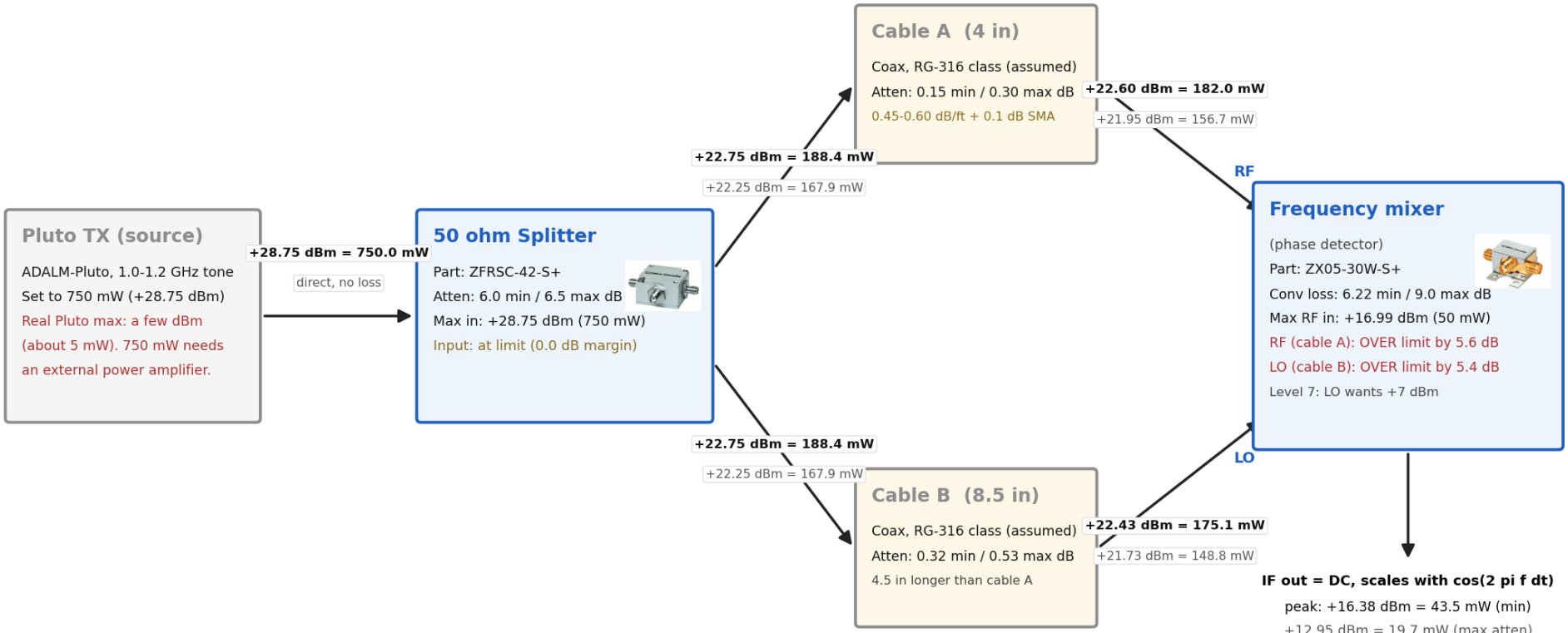


Same layout as racon_draft_design01b_drawing.pdf. Green text in each block says what that stage does. Each block also carries: title, part number, maximum input in dBm and mW, and minimum attenuation. Blue blocks are the nine parts we have, with datasheet maximum ratings and the photo from the datasheet; grey blocks are candidate parts named on the drawing and are not datasheet-verified. The small box beside each block is the running level relative to the start of that chain: worst case (maximum attenuation, minimum gain) to best case (minimum attenuation, maximum gain). Green means every term is datasheet-verified, amber means the total includes an assumed value.

3b. Bench test: Pluto TX to the phase detector

Bench test: Pluto TX -> splitter -> 4 in / 8.5 in cables -> frequency mixer (phase detector)

Source set to 750 mW (+28.75 dBm). On each arrow: level entering the next block with minimum attenuation (bold) and maximum attenuation (grey).



What this means at 750 mW

Splitter input is exactly on its 0.75 W damage rating (no margin).

Both mixer ports get ~+22 dBm (~175 mW), 3.5x its 50 mW max: it would be damaged.

Safe source levels

Max without damaging the mixer: +23.14 dBm = 206.0 mW
For the mixer's nominal +7 dBm LO: +13.32 dBm = 21.5 mW

Splitter and mixer values: datasheets (see the power-ratings doc).
Cable loss is ASSUMED - swap in your cable's datasheet values.

Pluto TX -> ZFRSC-42-S+ splitter -> 4 in and 8.5 in cables -> ZX05-30W-S+ mixer used as a phase detector, for a 750 mW (+28.75 dBm) source. Splitter and mixer attenuation and limits are datasheet values; cable loss is assumed for RG-316-class coax. A real ADALM-Pluto cannot deliver 750 mW, and at that level both mixer ports would be about 5.5 dB over the 50 mW rating.

4. Headroom against the damage limits

Expected levels come from the LTspice model in simulations/claude/01_jacon_system. Worst case is the -10 dBm close-range echo.

Part	Expected worst-case input	Damage limit	Margin	Action
ZX60-5123LN+ (LNA)	limiter output, +16.3 dBm while the limiter clamps	+15 dBm	**OVER by 1.3 dB**	Keep the in-line attenuator the drawing calls for.
ZX05-14LH-S+ (RX mixer)	LNA saturated output, about +17 to +18 dBm	+17 dBm (50 mW)	**0 to -1 dB**	Marginal at close range; the same pad fixes this.
ZX60-P162LN+ (IF amp)	mixer IF output, about +11 dBm worst case	+25 dBm	14 dB	Fine.
ZX60-P103LN+ (IF amp)	mixer IF output, about +11 dBm worst case	+21 dBm	10 dB	Fine.
ZX60-2534MA-S+ (IF amp)	mixer IF output, about +11 dBm worst case	-15 dBm	**OVER by 26 dB**	Do not put this straight after the mixer.
ZFRSC-42-S+ (both units)	IF amp output, about +20 dBm saturated	+28.8 dBm (0.75 W)	9 dB	Fine.
ZX05-30W-S+ (phase det.)	limiting-amp chain, about +8 dBm per port	+17 dBm (50 mW)	9 dB	Above the +1 dBm compression point, which is intended here.
ZX60-14012L-S+ (LO buffer)	PLL output, about +3 dBm	+10 dBm	7 dB	Check your ADF5355 output level; +3 dBm is an assumption.
ZX10-2-1252-S+ (LO split)	buffer output, about +14 dBm	+32.7 dBm (1.85 W)	19 dB	Fine.

5. Datasheet evidence, part by part

Each screenshot below is cropped straight out of the linked PDF, at the page stated in its caption. Open the PDF, go to that page, and the same table is there.

ZX60-5123LN+ - RX LNA

Datasheet: ZX60-5123LN+.pdf (Mini-Circuits) | local copy: datasheets/ZX60-5123LN+.pdf | product page: minicircuits.com

Parameter	Value
Band (datasheet)	5.5 - 12.5 GHz
Frequency in JACON	9.3 - 9.5 GHz
Gain / loss	+22.0 dB typ (19.0 min), 5.5-12.5 GHz
Damage limit	RF Input Power (CW) +15 dBm = +15.00 dBm = 31.62 mW (page 3)
Linear limit	+17.0 dBm typ output P1dB
Notes	Total power dissipation 1.2 W; DC +12 V

ABSOLUTE MAXIMUM RATINGS[2,3]

Parameter	Ratings
Operating Temperature (Case)	-40 °C to +85 °C
Storage Temperature	-55 °C to +100 °C
Total Power Dissipation[4]	1.2 W
RF Input Power (CW)	+15 dBm

ZX60-5123LN+ datasheet, page 3 - ABSOLUTE MAXIMUM RATINGS: RF Input Power (CW) = +15 dBm

ELECTRICAL SPECIFICATIONS AT +25 °C BASEPLATE AND V_S = +6V , UNLESS NOTED OTHERWISE

Parameter	Condition (GHz)	Min.	Typ.	Max.	Units
Frequency Range	-	5.5	-	12.5	GHz
Gain	5.5 — 12.5	19.0	22.0	-	dB
Gain Flatness	5.5 — 12.5	-	±0.6	-	dB
Input Return Loss	5.5 — 12.5	-	10.5	-	dB
Output Return Loss	5.5 — 12.5	-	14.0	-	dB
Directivity	5.5 — 12.5	-	27.0	-	dB
Output Power at 1 dB Compression (P1dB)	5.5 — 12.5	-	+17.0	-	dBm
Output Third Order Intercept Point (Output Power = 0 dBm)	5.5 — 12.5	-	+29.5	-	dBm

ZX60-5123LN+ datasheet, page 1 - ELECTRICAL SPECIFICATIONS: gain 22.0 dB typ, P1dB +17.0 dBm typ

ZX05-14LH-S+ - RX down-mixer / TX up-mixer

Datasheet: ZX05-14LH-S+.pdf (Mini-Circuits) | local copy: datasheets/ZX05-14LH-S+.pdf | product page: minicircuits.com

Parameter	Value
Band (datasheet)	RF+LO 3.7 - 10 GHz, IF DC - 4 GHz
Frequency in JACON	RF 9.3-9.5 GHz, LO 8.3 GHz, IF 1.0-1.2 GHz
Gain / loss	conv. loss 7.72 dB typ at RF 9400 MHz (6.7 dB band typ), LO +10 dBm
Damage limit	RF Power 50 mW = +16.99 dBm = 50 mW (page 1)
Linear limit	+5 dBm typ RF input at 1 dB compression
Notes	Level 10 mixer: LO drive +10 dBm

Maximum Ratings

Operating Temperature	-40°C to 85°C
Storage Temperature	-55°C to 100°C
RF Power	50mW

Permanent damage may occur if any of these limits are exceeded.

Coaxial Connections

LO	2
RF	3
IF	1

Features

- wide bandwidth, 3700 to 10
- low conversion loss, 6.7 dB
- high L-R isolation, 38 dB ty
- excellent IF BW, DC to 400
- rugged construction
- small size
- useable as up and down co
- protected by US patents 6,7

Applications

- satellite up and down conve
- defense radar and commun
- line of sight links
- federal fixed service

ZX05-14LH-S+ datasheet, page 1 - Maximum Ratings: RF Power = 50 mW

(MHz)		(dB)			(dB)		(dB)		a
LO/RF f_L-f_U	IF	Typ.	σ	Max.	Typ.	Min.	Typ.	Min.	
3700-10000	**DC-4000**								
3700-6200		6.7	0.3	8.3	38	30	17	11	
6200-10000		6.7	0.2	9.5	35	25	17	9	

1 dB COMPR.: +5 dBm typ.
* Conversion loss at 30 MHz IF. σ is a measure of repeatability from unit to unit.

ZX05-14LH-S+ datasheet, page 1 - 1 dB COMPR.: +5 dBm typ (LO +10 dBm)

Typical Performance Data

Frequency (MHz)		Conversion Loss (dB)	Isolation L-R (dB)	Isolation L-I (dB)	VSWR RF Port (:1)	VSWR LO Port (:1)
RF	LO	LO +10dBm	LO +10dBm	LO +10dBm	LO +10dBm	LO +10dBm
3700.10	3730.10	6.39	48.11	18.94	3.12	8.31
4100.10	4130.10	6.13	39.12	18.68	3.09	5.19
4500.10	4530.10	5.86	41.21	16.16	2.56	3.25
4900.10	4930.10	6.09	38.80	14.56	2.77	2.19
5300.10	5330.10	6.57	36.16	13.89	3.74	1.41
5700.10	5730.10	6.38	34.89	14.71	3.39	1.73
6100.10	6130.10	6.03	37.29	16.01	2.95	2.49
6200.10	6230.10	6.00	36.21	16.24	2.84	2.73
6600.10	6630.10	6.07	36.65	16.84	2.64	3.60
7000.10	7030.10	6.00	37.63	16.76	2.13	3.91
7400.10	7430.10	5.64	35.86	14.97	1.68	3.92
7800.10	7830.10	5.52	32.87	13.26	1.56	3.35
8200.10	8230.10	5.45	31.01	13.24	1.84	2.18
8600.10	8630.10	6.13	35.65	16.37	2.58	1.81
9000.10	9030.10	6.88	32.71	21.25	3.48	1.85
9400.10	9430.10	7.72	27.89	19.20	4.01	2.05
9700.10	9730.10	8.26	27.01	17.18	4.80	2.13

ZX05-14LH-S+ datasheet, page 1 - Typical Performance Data: conversion loss 7.72 dB at RF 9400 MHz (band typ is 6.7 dB)

ZX60-14012L-S+ - LO buffer amplifier

Datasheet: ZX60-14012L-S+.pdf (Mini-Circuits) | local copy: datasheets/ZX60-14012L-S+.pdf | product page: minicircuits.com

Parameter	Value
Band (datasheet)	0.3 MHz - 14 GHz
Frequency in JACON	8.3 GHz

Gain / loss	12.55 dB at 8.0 GHz, 11.85 dB at 9.0 GHz -> approx 12.3 dB at 8.3 GHz
Damage limit	Input Power (no Damage) 10 dBm = +10.00 dBm = 10 mW (page 1)
Linear limit	+11 dBm typ output P1dB (+4 dBm min at 10-14 GHz)
Notes	Band-wide spec is +12 dB typ / 9 dB min; the page 2 table gives 12.55 dB at 8 GHz. NF 5.35 dB at 8 GHz

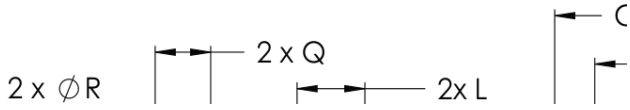
2. Unit is internally voltage regulated over 10.5 to 20 VDC input range.
3. +4 dBm min. power output @ 10000-14000 MHz

Maximum Ratings

Operating Temperature	-40°C to 80°C case
	-40°C to 55°C ambient at 12V
	-40°C to 50°C ambient at 15V
	-40°C to 40°C ambient at 20V
Storage Temperature	-55°C to 100°C
DC Voltage	20V
Input Power(no Damage)	10dBm

Permanent damage may occur if any of these limits are exceeded.

Outline Drawing



ZX60-14012L-S+ datasheet, page 1 - Maximum Ratings: Input Power (no Damage) = 10 dBm

Electrical Specifications at T_{AMB} = 25°C

MODEL NO.	FREQ. (MHz)	GAIN (dB)			MAXIMUM POWER (dBm)	DYNAMIC RANGE		VSWR (:1) Typ.		ACTIVE DIRECTIVITY (dB)	DC VOLTAGE @ Pin V+ (V) (Note 2)	DC OPERATING CURRENT @ Pin V+ (mA)	
				Flatness	Output (1 dB Comp.)	NF (dB) (Note 1)	IP3 (dBm)			Isolation-Gain			
	f_L - f_U	Typ.	Min.	Typ. Max.	Typ. Min. (Note 3)	Typ. Max.	Typ.	In	Out	Typ.		Typ.	Max.

ZX60-14012L-S+ datasheet, page 1 - Electrical Specifications: gain 12 dB typ, output P1dB +11 dBm typ

FREQUENCY (MHz)	GAIN (dB)	DIRECTIVITY (dB)	VSWR IN (:1)	VSWR OUT (:1)	POWER OUT @1dB COMPRESSION (dBm)	IP3 (dBm)	NF (dB)
0.3	11.68	8.13	1.26	1.96	10.56	24.86	------
10	11.29	7.89	1.19	1.72	10.99	25.44	5.21
500	11.31	7.86	1.21	1.71	10.88	25.08	4.83
750	10.99	8.57	1.21	1.67	11.03	25.55	5.03
1000	11.03	8.23	1.20	1.57	11.54	26.40	5.04
2000	11.72	7.03	1.26	1.51	12.37	28.02	4.72
3000	11.94	6.45	1.22	1.35	12.01	27.22	4.79
4000	12.87	4.67	1.14	1.24	13.22	25.58	4.93
5000	12.86	4.55	1.19	1.47	13.04	22.99	4.90
6000	12.71	4.76	1.24	1.67	13.04	21.14	5.00
7000	12.45	5.00	1.20	1.67	13.03	20.91	5.02
8000	12.55	5.35	1.34	1.65	12.25	17.80	5.23
9000	11.85	5.87	1.62	1.72	11.18	16.77	5.35
10000	11.98	5.84	1.81	1.77	10.57	14.99	5.75

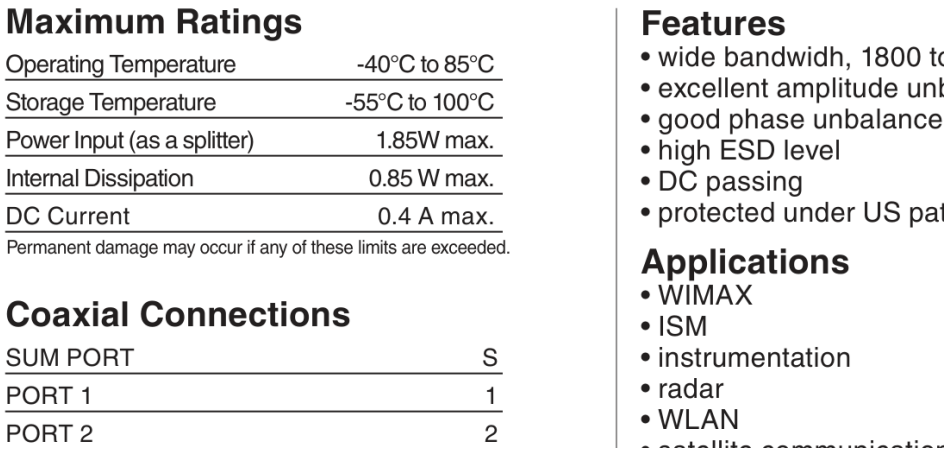
ZX60-14012L-S+ datasheet, page 2 - Typical Performance Data: 12.55 dB gain at 8000 MHz, 11.85 dB at 9000 MHz (8.3 GHz falls between)

ZX10-2-1252-S+ - LO 2-way splitter (DC pass)

Datasheet: ZX10-2-1252-S+.pdf (Mini-Circuits) | local copy: datasheets/ZX10-2-1252-S+.pdf | product page: minicircuits.com

Parameter	Value

Band (datasheet)	1800 - 12500 MHz
Frequency in JACON	8.3 GHz
Gain / loss	-3.0 dB split + 0.7 dB typ insertion loss
Damage limit	Power Input (as a splitter) 1.85 W = +32.67 dBm = 1850 mW (page 2)
Linear limit	passive - no compression spec
Notes	Internal dissipation 0.85 W max; DC current 0.4 A max



Maximum Ratings

Operating Temperature	-40°C to 85°C
Storage Temperature	-55°C to 100°C
Power Input (as a splitter)	1.85W max.
Internal Dissipation	0.85 W max.
DC Current	0.4 A max.

Permanent damage may occur if any of these limits are exceeded.

Coaxial Connections

SUM PORT	S
PORT 1	1
PORT 2	2

Features

- wide bandwidh, 1800 t
- excellent amplitude unb
- good phase unbalance
- high ESD level
- DC passing
- protected under US pat

Applications

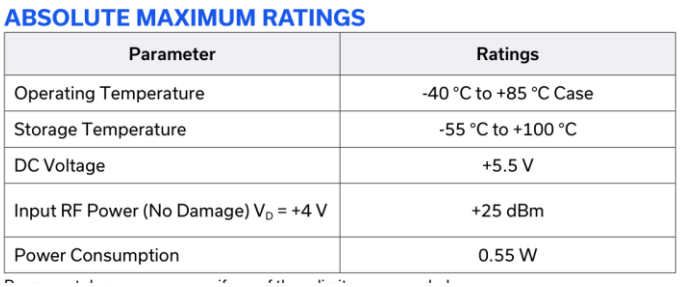
- WIMAX
- ISM
- instrumentation
- radar
- WLAN

ZX10-2-1252-S+ datasheet, page 2 - Maximum Ratings: Power Input (as a splitter) = 1.85 W

ZX60-P162LN+ - IF amplifier (preferred for 1.0-1.2 GHz)

Datasheet: ZX60-P162LN+.pdf (Mini-Circuits) | local copy: datasheets/ZX60-P162LN+.pdf | product page: minicircuits.com

Parameter	Value
Band (datasheet)	0.7 - 1.6 GHz
Frequency in JACON	1.0 - 1.2 GHz
Gain / loss	21.81 dB at 1.1 GHz (22.49 at 1.0, 21.12 at 1.2 GHz)
Damage limit	Input RF Power (No Damage) VD=+4 V: +25 dBm = +25.00 dBm = 316.2 mW (page 2)
Linear limit	+19.9 dBm typ output P1dB at 1.0 GHz (+17.5 min)
Notes	Noise figure 0.52 dB typ at 1.0 GHz; power consumption 0.55 W



ABSOLUTE MAXIMUM RATINGS

Parameter	Ratings
Operating Temperature	-40 °C to +85 °C Case
Storage Temperature	-55 °C to +100 °C
DC Voltage	+5.5 V
Input RF Power (No Damage) V_D = +4 V	+25 dBm
Power Consumption	0.55 W

ZX60-P162LN+ datasheet, page 2 - ABSOLUTE MAXIMUM RATINGS: Input RF Power (No Damage) = +25 dBm

ELECTRICAL SPECIFICATIONS AT +25 °C AND +4.0 V, UNLESS NOTED OTHERWISE

Parameter	Condition (GHz)	Min.	Typ.	Max.	Units
Frequency Range		0.7		1.6	GHz
Noise Figure	0.7		0.65	0.95	dB
	0.8		0.47		
	1.0		0.52		
	1.3		0.56		
	1.6		0.74		
Gain	0.7	20.9	24.1	24.5	dB
	0.8		23.8		
	1.0		22.5		
	1.3		20.5		
	1.6		18.5		
Output Power at 1 dB Compression	0.7	+17.5	+19.2		dBm
	0.8		+19.9		
	1.0		+19.9		
	1.3		+19.7		
	1.6		+19.5		
Output IP3	0.7	+28.0	+29.0		dBm
	0.8		+29.8		
	1.0		+29.9		
	1.3		+30.2		

ZX60-P162LN+ datasheet, page 2 - ELECTRICAL SPECIFICATIONS: gain and output P1dB vs frequency

Frequency (MHz)	Gain (dB)	Directivity (dB)	VSWR (:1) +4 V		Power Out at 1 dB COMPR. (dBm)	Noise Figure (dB)	Output IP3 (dBm)
	+4 V	+4 V	IN	OUT	+4 V	+4 V	+4 V
700	24.09	13.78	2.19	1.58	19.2	0.7	29.0
800	23.78	13.74	1.64	1.43	19.9	0.5	29.8
900	23.17	14.08	1.35	1.37	20.0	0.6	30.2
1000	22.49	14.69	1.19	1.38	19.9	0.5	29.9
1100	21.81	15.35	1.09	1.44	19.9	0.6	30.3
1200	21.12	16.12	1.16	1.53	20.2	0.6	30.4

ZX60-P162LN+ datasheet, page 4 - Typical Performance Data: 21.81 dB gain and +19.9 dBm P1dB at 1100 MHz

ZX60-P103LN+ - IF amplifier (wideband alternate)

Datasheet: ZX60-P103LN+.pdf (Mini-Circuits) | local copy: datasheets/ZX60-P103LN+.pdf | product page: minicircuits.com

Parameter	Value
Band (datasheet)	50 - 3000 MHz
Frequency in JACON	1.0 - 1.2 GHz
Gain / loss	+15.6 dB typ at 1000 MHz (13.5 min)
Damage limit	Input RF Power (no damage): 21 dBm (50-2000 MHz), 26 dBm (2000-3000 MHz) = +21.00 dBm = 125.9 mW (page 3)
Linear limit	+22.4 dBm typ output P1dB at 1000 MHz
Notes	Noise figure 0.5 dB typ at 1000 MHz; power consumption 0.66 W

ABSOLUTE MAXIMUM RATINGS

Parameter	Ratings
Operating Temperature	-40°C to 85°C Case
Storage Temperature	-55°C to 100°C
DC Voltage	+5.5V
Input RF Power (no damage)	50 MHz - 2000 MHz 21dBm 2000 MHz- 3000 MHz 26dBm
Power Consumption	0.66W

Permanent damage may occur if any of these limits are exceeded.

ZX60-P103LN+ datasheet, page 3 - ABSOLUTE MAXIMUM RATINGS: Input RF Power (no damage) = 21 dBm at 50-2000 MHz

ELECTRICAL SPECIFICATIONS AT +25°C

Parameter	Condition	Min.	Typ.	Max.	Units
Frequency Range		50		3000	MHz
Noise Figure	50		1.2		dB
	500		0.4		
	1000		0.5		
	2000		0.6		
	3000		1.0		
Gain	50	23.0	25.2		dB
	500	18.0	20.3		
	1000	13.5	15.6		
	2000	8.0	10.0		
	3000	4.5	6.9		
Output Power at 1dB Compression	50		19.8		dBm
	500		22.3		
	1000		22.4		
	2000		23.2		
	3000		23.8		
Output IP3	50		36.9		dBm
	500		39.7		
	1000		39.4		
	2000		42.6		

ZX60-P103LN+ datasheet, page 2 - ELECTRICAL SPECIFICATIONS: gain 15.6 dB typ and P1dB +22.4 dBm typ at 1000 MHz

ZX60-2534MA-S+ - IF amplifier (high gain alternate)

Datasheet: ZX60-2534MA-S+.pdf (Mini-Circuits) | local copy: datasheets/ZX60-2534MA-S+.pdf | product page: minicircuits.com

Parameter	Value
Band (datasheet)	0.5 - 2.5 GHz
Frequency in JACON	1.0 - 1.2 GHz
Gain / loss	+43.5 dB typ at 1.0 GHz (at VD = 5 V)
Damage limit	Input Power (no damage) -15 dBm = -15.00 dBm = 0.03162 mW (page 1)
Linear limit	+19.0 dBm typ output P1dB at fL
Notes	LOWEST damage limit of any part here. Power dissipation 0.5 W; DC 5 V

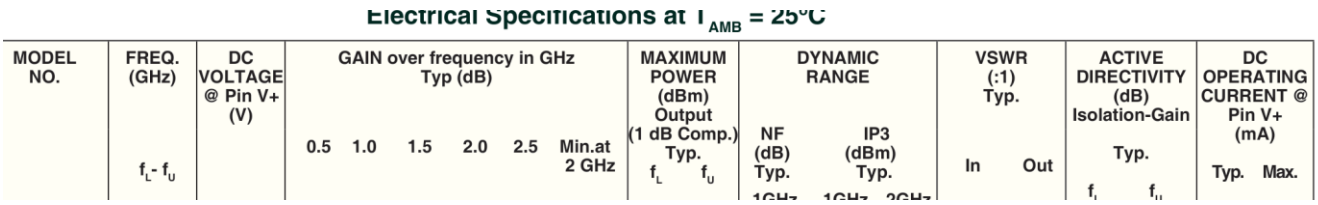
Maximum Ratings

Operating Temperature	-40°C to 80°C case
Storage Temperature	-55°C to 100°C
DC Voltage	7V
Input Power (no damage)	-15dBm
Power Dissipation	0.5W

Permanent damage may occur if any of these limits are exceeded.

Outline Drawing

ZX60-2534MA-S+ datasheet, page 1 - Maximum Ratings: Input Power (no damage) = -15 dBm



Electrical Specifications at T_{AMB} = 25°C

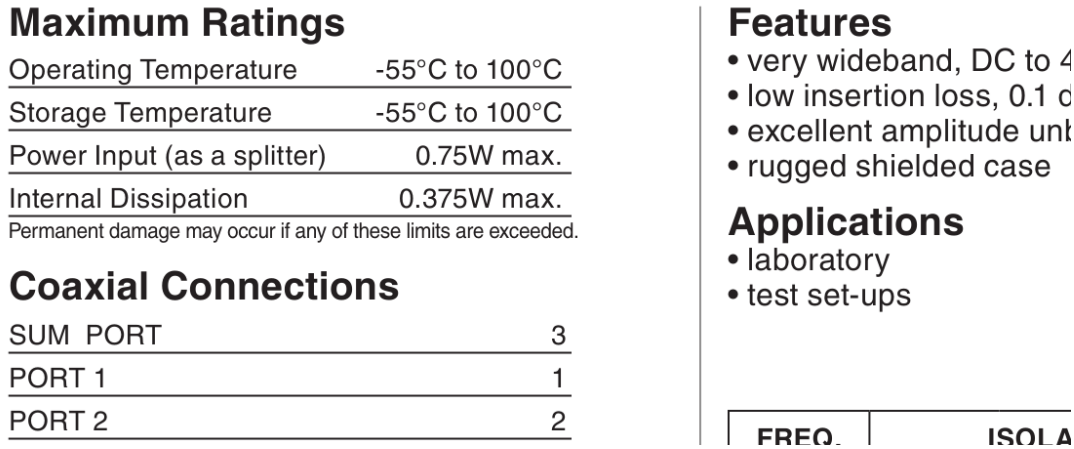
MODEL NO.	FREQ. (GHz) f_L - f_U	DC VOLTAGE @ Pin V+ (V)	GAIN over frequency in GHz Typ (dB) 0.5	1.0	1.5	2.0	2.5	Min.at 2 GHz	MAXIMUM POWER (dBm) Output (1 dB Comp.) Typ. f_L	f_U	DYNAMIC RANGE NF (dB) Typ.	IP3 (dBm) Typ.	VSWR (:1) Typ. In	Out	ACTIVE DIRECTIVITY (dB) Isolation-Gain Typ. f_L	f_U	DC OPERATING CURRENT @ Pin V+ (mA) Typ.	Max.
											1GHz	1GHz 2GHz						

ZX60-2534MA-S+ datasheet, page 1 - Electrical Specifications: gain over frequency, output P1dB

ZFRSC-42-S+ - Resistive 2-way splitter (IF split and delay-line split)

Datasheet: ZFRSC-42-S+.pdf (Mini-Circuits) | local copy: datasheets/ZFRSC-42-S+.pdf | product page: minicircuits.com

Parameter	Value
Band (datasheet)	DC - 4200 MHz
Frequency in JACON	1.0 - 1.2 GHz
Gain / loss	-6.0 dB total loss (5.99 / 6.02 dB at 1000 MHz)
Damage limit	Power Input (as a splitter) 0.75 W = +28.75 dBm = 750 mW (page 1)
Linear limit	passive - no compression spec
Notes	Internal dissipation 0.375 W max. Resistive: isolation equals insertion loss (~6 dB)



Maximum Ratings

Operating Temperature	-55°C to 100°C
Storage Temperature	-55°C to 100°C
Power Input (as a splitter)	0.75W max.
Internal Dissipation	0.375W max.

Permanent damage may occur if any of these limits are exceeded.

Coaxial Connections

SUM PORT	3
PORT 1	1
PORT 2	2

Features

- very wideband, DC to 4
- low insertion loss, 0.1 d
- excellent amplitude unb
- rugged shielded case

Applications

- laboratory
- test set-ups

FREQ. ISOLA

ZFRSC-42-S+ datasheet, page 1 - Maximum Ratings: Power Input (as a splitter) = 0.75 W, internal dissipation 0.375 W

ZX05-30W-S+ - Phase detector (delay-line mixer)

Datasheet: ZX05-30W-S+.pdf (Mini-Circuits) | local copy: datasheets/ZX05-30W-S+.pdf | product page: minicircuits.com

Parameter	Value
Band (datasheet)	RF+LO 300 - 4000 MHz, IF DC - 950 MHz
Frequency in JACON	RF+LO 1.0-1.2 GHz, IF = DC
Gain / loss	conv. loss 6.22 dB typ at RF 944 MHz (6.8 dB band typ), LO +7 dBm
Damage limit	RF Power 50 mW = +16.99 dBm = 50 mW (page 1)
Linear limit	+1 dBm typ RF input at 1 dB compression
Notes	Level 7 mixer: LO drive +7 dBm. IF current 40 mA max

Maximum Ratings

Operating Temperature	-40°C to 85°C
Storage Temperature	-55°C to 100°C
RF Power	50mW
IF Current	40mA

Permanent damage may occur if any of these limits are exceeded.

Coaxial Connections

LO	1
RF	2
IF	3

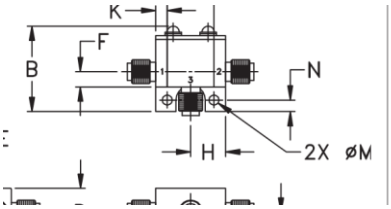
Features

- rugged construction
- small size
- low conversion loss
- high L-R isolation
- protected by US Patents

Applications

- cellular
- PCS
- instrumentation
- satellite communication

ZX05-30W-S+ datasheet, page 1 - Maximum Ratings: RF Power = 50 mW, IF Current = 40 mA



f_L-f_U		$\overline{X}$	σ	Max.	Max.	Typ.	Min.	Typ.	Min.
300-4000	DC-950	6.8	0.2	9.0	9.8	35	17	16	7

1 dB COMP.: +1 dBm typ.
m= mid band [$2f_L$ to $f_U/2$]

Typical Performance Data

Frequency	Conversion	Isolation	Isolation	VSWR	VS

ZX05-30W-S+ datasheet, page 1 - 1 dB COMP.: +1 dBm typ; conversion loss 6.8 dB typ at LO +7 dBm

Typical Performance Data

Frequency (MHz)		Conversion Loss (dB)	Isolation L-R (dB)	Isolation L-I (dB)	VSWR RF Port (:1)	VSWR LO Port (:1)
RF	LO	LO +7dBm	LO +7dBm	LO +7dBm	LO +7dBm	LO +7dBm
300.10	330.10	5.34	45.30	34.86	2.04	5.44
450.10	480.10	4.98	41.41	32.02	1.54	3.61
600.10	630.10	5.02	39.35	28.89	1.73	3.17
781.35	751.35	5.11	37.95	28.09	2.81	2.51
943.85	913.85	6.22	37.02	28.05	3.48	1.97
1106.35	1076.35	6.36	39.38	23.65	2.77	1.70
1268.85	1238.85	6.45	42.76	20.82	2.59	1.80

ZX05-30W-S+ datasheet, page 1 - Typical Performance Data: conversion loss 6.22 dB at RF 944 MHz, closest tabulated point to our 1.0-1.2 GHz

6. What is assumed, and what to check yourself

- Levels in the flow chart come from the LTspice model, not from measurements. They assume a -30 dBm echo, a +3 dBm ADF5355 LO output, and the limiter / limiting-amp behaviour described in the drawing.
- The ZX60-14012L-S+ gain (+12 dB typ) and P1dB (+11 dBm typ) are the band-wide specs for 0.3 MHz to 14 GHz. For 8.3 GHz specifically, read the typical performance curves on page 2 of its datasheet.
- Mini-Circuits quotes mixer damage as RF Power 50 mW. That is the RF port limit; the LO drive level is a separate specification (+10 dBm for the ZX05-14LH-S+, +7 dBm for the ZX05-30W-S+).
- The ZX60-2534MA-S+ datasheet is published as ZX60-2534MA+.pdf and covers both the connectorised -S+ version and the bare model; the model table on page 1 lists ZX60-2534MA-S+.

- The power limiter, limiting amplifier, envelope detector, VCO, keying switch, image filter and power amp are not among the nine parts, so they are not datasheet-verified in this document.
- Gain and loss figures are typical values at room temperature. Minimum / maximum columns on the datasheets are what you should design against.
- The simulated levels were computed with the band-typical 6.7 dB mixer conversion loss. The ZX05-14LH-S+ typical performance table gives 7.72 dB at RF 9400 MHz, so expect the real IF to be about 1 dB lower than the red numbers in the flow chart.
- Gain and conversion-loss values are now quoted at the frequency each part actually sees, taken from the typical performance tables, not the band-wide headline figures. Both are shown where they differ.